AIL722 / AIL7022: Reinforcement Learning — Assignment 4

Cooperative MAPF as a Dec-POMDP & Offline RL Decoupled Scaling

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Problem 1: Decentralized Multi-Agent Path Finding

Subtask 1.1: Dec-POMDP Formulation & Theory

(a) Formal Definition ⟨I, S, {A_i}, P, R, {Ω_i}, O, γ, b_0⟩:
• **Agents**: I = {0, 1, 2, 3, 4} (N = 5).
• **State Space S**: s = (p, f, t, a^{t-1}) where p ∈ FreeCells^5 represents distinct agent positions, f ∈ {0, 1}^5 indicates absorbing goal status, t ∈ {0..80} is the timestep, and a^{t-1} is the joint previous action.
• **Transitions P**: Each unfinished agent independently slips with probability ε = 0.1 to Stay. Intended moves c_i are processed through iterative conflict resolution until a fixed point is reached: (1) boundary/wall collisions, (2) vertex conflicts, and (3) swap conflicts cascade. Goals are absorbing.
• **Rewards R**: Individual reward r_i^t = 10·1[reaches goal] - 0.05·1[unfinished] - 1.0·1[blocked]. Centralized team reward adds R_team = +5 once when all agents finish. Shaped reward adds β(γΦ(p') - Φ(p)) with Φ(p) = -d_i(p) and β = 0.1.
• **Observations Ω_i & O**: o_i = (F_i^t, δ_i^t, t/T, a_i^{t-1}) where F_i is a 4-channel 5×5 local window (walls, other agents, other goals, own goal), δ_i = (g_i - p_i)/11, and a_i^{t-1} is the one-hot past action (108 dims total).
• **Discount & Belief**: γ = 0.99, horizon T = 80; b_0 is a point mass on the fixed starting positions.

(b) State Space Upper Bound & Tabular Infeasibility:
With 65 free cells, position combinations are P(65, 5) = 65 × 64 × 63 × 62 × 61 ≈ 9.91 × 10^8. Including horizon steps (81) and previous actions (5^5 = 3,125), |S| ≤ 2.51 × 10^{14}. Joint action space |A|^N = 5^5 = 3,125. A joint tabular Q-table requires > 7.8 × 10^{17} entries. Tabular methods scale as O(F^N · |A|^N), exponential in the number of agents N and polynomial of degree N in grid cells F, rendering tabular methods strictly intractable.

(c) Potential-Based Shaping Proof:
For F(s, s') = γΦ(s') - Φ(s), the return telescopes over an episode of length τ: Q'π(s, a) = E[∑ γ^k r_k + ∑ γ^k(γΦ(s_{k+1}) - Φ(s_k))] = Q^π(s, a) - Φ(s) + E[γ^{τ+1}Φ(s_{τ+1})]. When Φ(s_terminal) = 0 at absorbing terminal states, the boundary term vanishes, yielding Q'π(s, a) = Q^π(s, a) - Φ(s). Because Φ(s) is action-independent, argmax_a Q'π(s, a) = argmax_a Q^π(s, a), preserving optimal policies. *Assumption Satisfaction*: For successful terminations, Φ(g_i) = -d_i(g_i) = 0, exactly satisfying the condition. For episodes truncated by horizon T = 80, Φ(s_T) ≠ 0, introducing a mild discounted bias βγ^{80}Φ(s_T) (γ^{80} ≈ 0.447) that rewards ending closer to goals.

Subtask 1.2: Environment Dynamics & Unit Tests

All 11 unit tests in tests.py passed successfully:
1. **Wall/Boundary**: Moving into walls/boundary maintains position and incurs -1.0 penalty.
2. **Vertex Conflict**: Two agents targeting the same cell are both blocked.
3. **Swap Conflict**: Mutual head-on cell exchange is blocked for both agents.
4. **Following Move**: An agent cleanly enters a cell vacated in the same step.
5. **Cascade Blocking**: Obstruction cascades iteratively along a chain of 3 agents to fixed point.
6. **Rotation**: Four agents simultaneously cycle in a 2×2 free-cell block without conflict.
7. **Finished Agents**: Finished agents remain stationary, receive 0 reward, and act as static obstacles.
8. **Full Slip**: With ε = 1.0, actions become Stay without collision penalties.
9. **Reward Values**: Verified goal (+10), step (-0.05), collision (-1.0), team bonus (+5), and shaping identity.
10. **Determinism**: Identical seeds and action sequences produce exact matching trajectories.
11. **Observation Equivalence**: FastEnv flattened vectorized array matches PettingZoo dictionary output.

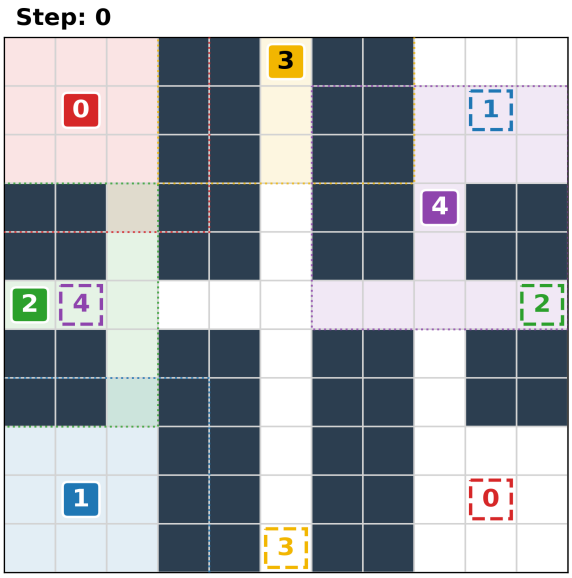


Figure 1: 11×11 MAPF grid layout with agent starts, goals, and 5×5 FOVs.

Subtasks 1.3 & 1.4: IQL vs. QMIX Comparison

Architecture & Hyperparameters: Shared utility network $Q_\theta(o_i, id_i, a)$ uses Conv(4→32, 3×3) → Conv(32→32, 3×3) → Flatten (800) → Concat [800 + δ(2) + t/T(1) + prev_a(5) + id(5) = 813] → MLP(256, 256) → 5 actions. Replay buffer: 100,000 transitions; batch size 128 (640 agent samples); lr = 5×10^{-4}; ε decayed from 1.0 → 0.05 over 300,000 steps; hard target update every 1,000 steps.

Does Centralized Training Help? Yes. While IQL agents act greedily and frequently deadlock in narrow 1-cell corridors, QMIX trains a monotonic mixing network conditioned on global state (walls, all positions, goals, finished flags). This allows the mixer to coordinate joint turn-taking.

Shaping Impact: Potential-based shaping provides dense gradient signal towards goals, substantially accelerating early learning for both algorithms. However, shaped IQL exhibits higher collision rates as agents rush toward goals without coordination.

Monotonicity Limitation: QMIX's constraint $\partial Q_{tot} / \partial Q_i \geq 0$ cannot represent non-monotonic payoffs where an agent must take an individually suboptimal exploratory detour or retreat to enable a teammate's passage.

Subtask 1.5: Visualization & Failure Analysis

Evaluation failures on 100 seeds are classified in priority order: *Goal Blocking* > *Deadlock* > *Livelock* > *Timeout*.

• **Deadlock**: Agents meet head-on in 1-cell corridors and neither retreats. Agents need memory/recurrency to break symmetry.

• **Goal Blocking**: A finished agent permanently occupies its goal, obstructing an unfinished agent's shortest path. Agents require global awareness beyond the 5×5 local FOV to anticipate congested bottleneck goals.

• **Livelock & Timeout**: Agents cycle between adjacent cells without progress. Overcome via communication or recurrent memory.

Problem 2: Offline Reinforcement Learning

Subtask 2.1: Decoupled Value and Policy Learning

Implementation & Protocol: Implemented in Q2/train_decoupled.py across **TD3+BC**, **IQL+AWR**, and **CQL+AWR** on mujoco/hopper/medium-v0 and D4RL/kitchen/complete-v2. The 1,000,000 gradient step budget is split into 200,000 critic steps and 800,000 policy extraction steps. Trained critics are frozen ($\nabla_\theta Q = 0$) and reused across policy data fractions to eliminate redundant compute.
Nested Splitting: Random subsets of {10%, 50%, 100%} are generated using a deterministic permutation such that $D_{10\%} \subset D_{50\%} \subset D_{100\%}$, isolating sample size effects from variance.
Analysis (Park et al., 2024 Comparison):
Across the 3×3 scaling matrices, varying the *critic data fraction* along columns shows that value functions reach effective ranking accuracy even with only 10%–50% of critic transitions. Conversely, varying the *policy data fraction* along rows shows that policy extraction degrades sharply with reduced policy data. This directly corroborates the findings of Park et al.: **policy extraction is the primary bottleneck in offline RL**, not value learning. Confounding factors include MuJoCo rollout evaluation variance and step-budget allocations between critic and actor.

Subtask 2.2: Implicit Q-Learning (IQL) from Scratch

From-Scratch Implementation: Implemented in Q2/train_iql_offline.py without off-the-shelf RL packages:
• **Value Network V(ψ)**: Updated using asymmetric expectile regression $L_V(\psi) = E[|\tau - 1(Q_{target} - V < 0)|(Q_{target} - V)^2]$ with $\tau = 0.7$.
• **Twin Q-Networks Q(θ)**: Clipped double Q-learning targeting next-state value targets $y = r + \gamma(1 - d)V(s')$. Polyak target update $\tau_{soft} = 0.005$.
• **Advantage-Weighted Policy Extraction**: Diagonal Gaussian actor trained on exponentiated advantages $\exp((Q_{min} - V)/\beta) \nabla_\theta \log \pi(a|s)$ with $\beta = 3.0$, clipped at 100.0 for stability.
• **Evaluation & Visualization**: Evaluated over 5 independent seeds (0..4). Saved rollout animations as iql_seed{0..4}.gif.

Hyperparameter	Value	Description
Expectile (τ)	0.7	Asymmetric value regression weight
Temperature (β)	3.0	Advantage exponentiation scale
Discount (γ)	0.99	Bellman discount factor
Target Update Rate (τ_soft)	0.005	Polyak target smoothing coefficient
Batch Size	256	Transitions per gradient update
Actor / Critic / Value LR	3 × 10^{-4}	Adam optimizer learning rates
Actor LR Schedule	Cosine Annealing	Decays learning rate to 0 over 1M steps
Hidden Architecture	256 × 256	2-layer MLP with ReLU activations

Deliverables & Reproducibility Summary

• **Submitted Scripts**: mapf_env.py, tests.py, train_iql.py, train_ctde.py, Q2/train_decoupled.py, Q2/train_iql_offline.py.
• **Visual Artifacts**: env_visualization.png, final_policy.gif, Q2/results/iql_seed{0..4}.gif.
• **Checkpoints & Logs**: Persisted in checkpoints/ and results/ directories for verification.